R1.2 Energy cycles in reactions

Bond breaking and forming

Learning outcomes

By the end of this section you should be able to:

- Calculate the enthalpy change of a reaction from given average bond enthalpy data.
- Describe why bond enthalpy data are average values and may differ from those measured experimentally.

Imagine two dogs playing tug-of-war with a toy. Both dogs are 'sharing' the toy and if enough force is applied, the toy could break. Like the 'shared' toy in **Figure 1**, a chemical bond between two atoms can break if enough energy is applied. Specifically, in the case of a bond that is breaking, energy must be absorbed. Where does this energy come from? We must remember that since energy is conserved in a closed system, it must have come from the surroundings. If breaking a bond requires energy, then forming a bond must release energy to the surroundings. Can you think of a way to remember this?

Study skills

Sometimes it is helpful to think of everyday experiences as examples for remembering concepts in chemistry. For bond formation you can think of this as two magnets that are attracted to each other. Imagine holding them a short distance away from each other and then allowing the magnets to 'snap' together. The sound of them hitting each other is the energy that is released, in much the same way that energy is released when bonds are formed.

Figure 1. Two dogs playing tug-of-war.

Credit: Bill Varie, Getty Images

Use **Interactive 1** to help you remember some of these important concepts.

Bond breaking	Bond forming

+ΔH	Energy is released	Endothermic	Energy is absorbed	−ΔH	Exothermic

Check

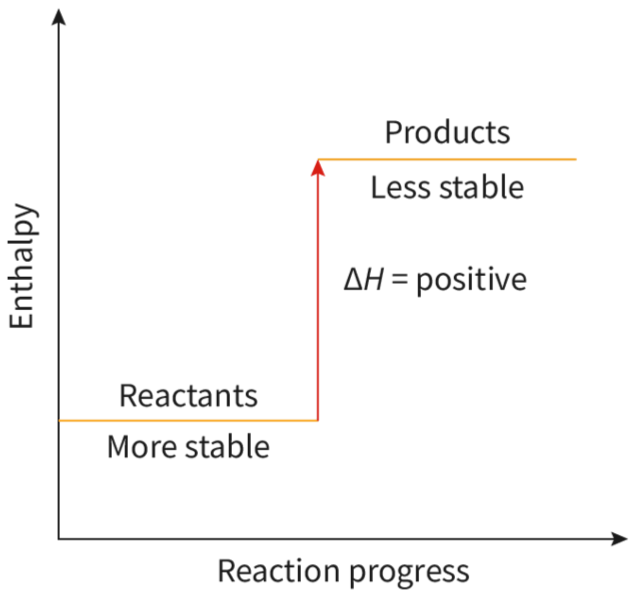
Interactive 1. Differentiating Between Bond Breaking and Formation.

Concept

Ionic bonds are formed by electrostatic attractions between oppositely charged ions and covalent bonds are formed by the electrostatic attraction between a shared pair of electrons and the positively charged nucleus. Energy must be absorbed to break these bonds and energy is released when forming bonds.

Bond enthalpy

When we look at the equation for a chemical reaction, sometimes we overlook the steps that must take place for the reactants to be transformed into the products. Each chemical reaction involves the breaking and forming of bonds, with an overall change in energy. This change in energy reveals important information about the stability of the reactants and products. Specifically, exothermic reactions have more stable products and endothermic reactions have less stable products compared to the reactants. This can be seen in the enthalpy diagrams in **Figure 2** where those chemical species with higher energy are less stable than those with lower energy. So, for an exothermic reaction, the products have less energy than the reactants and therefore the products are more stable (and vice versa for an endothermic reaction).



More information

The image is an enthalpy diagram representing an endothermic reaction. The X-axis is labeled 'Reaction progress' and the Y-axis is labeled 'Enthalpy'. The diagram depicts a reaction where the reactants are located at a lower enthalpy level, labeled as 'Reactants - More stable', indicating higher stability. An arrow points upward to a higher enthalpy level labeled 'Products - Less stable', reflecting lower stability. The change in enthalpy (ΔH) is marked as positive, showing the products have higher energy than the reactants.

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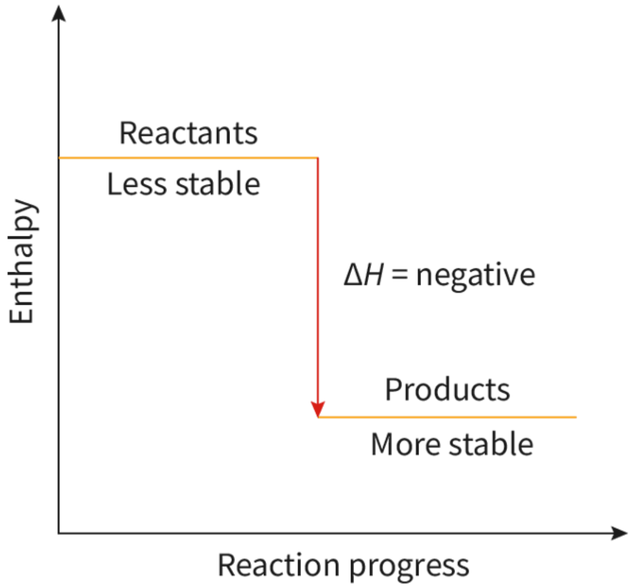

Figure 2. Endothermic and exothermic enthalpy diagrams comparing the stability of reactants and products.

More information for figure 2

This diagram is a graph representing the enthalpy change during a chemical reaction. The Y-axis is labeled 'Enthalpy' and the X-axis is labeled 'Reaction progress.' The graph shows two horizontal lines with the upper line labeled 'Reactants' signifying 'Less stable' and the lower line labeled 'Products' indicating 'More stable.' An arrow pointing downwards between the two lines is labeled 'ΔH = negative,' showing an exothermic reaction where the products are more stable than the reactants.

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Making connections

The relative stability of reactants and products determines whether reactions are endothermic or exothermic (see section R1.1.3 (https://app.kognity.com/:sectionlink:138726)).

If we consider this statement in relation to what we know about everyday molecules, we can reveal a useful method for remembering that bond breaking is endothermic and bond making is exothermic.

Consider the following reaction:

$$O_2 \rightarrow 2O$$

Kog

In this reaction, energy is required to break the double bond between the bonded oxygens, resulting in 2 oxygen atoms. But how does this relate to stability?

The form of oxygen that exists predominantly on Earth is diatomic oxygen and it is this way because this is the most stable form of oxygen. When the oxygen molecule is broken, the resulting oxygen atoms are considerably less stable. This matches **Figure 2** whereby O_2 must absorb energy in order to break the double bond.

To quantify the energy required to break a bond, we use bond enthalpy, *H* or bond dissociation energy, *E*. These terms can be used interchangeably and are defined as the energy required to break one mole of chemical bonds in the gaseous state. Bond enthalpy values are always positive because they refer to bonds being broken (bond breaking is endothermic).

Bond enthalpy values can be found in section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet. Examples of bond dissociation and formation are shown below. Use **Interactive 2** to investigate if they are endothermic or exothermic.

$Cl_2(g) \rightarrow 2Cl(g)$

Turn

Card 1 of 4

Interactive 2. Endothermic or Exothermic?

More information for interactive 2

An interactive carousel selector features four pluck cards, each displaying a chemical reaction. Users need to identify whether the given reaction is endothermic or exothermic.
Users can switch between cards using arrows present in the bottom right and bottom left. Users can check the answer by selecting the Turn button at the bottom of each card.
The reactions given in the card are as follows:
Card 1: $Cl_2(g) \rightarrow 2Cl\,(g)$
Card 2: $2F\,(g) \rightarrow F_2(g)$
Card 3: $2Cl\,(g) \rightarrow Cl_2(g)$
Card 4: $F_2(g) \rightarrow 2F\,(g)$

This interactivity helps users distinguish between exothermic and endothermic reactions.

Read below for the solution:

Card 1: Endothermic
$E(Cl–Cl) = +242\text{ kJ mol}^{-1}$

Card 2: Exothermic
$E(F–F) = -159\text{ kJ mol}^{-1}$

Card 3: Exothermic
$E(Cl–Cl) = -242\text{ kJ mol}^{-1}$

Card 4: Endothermic
$E(F–F) = +159\text{ kJ mol}^{-1}$

Importantly, you need to recognise that if these reactions were reversed so that they are for the bonds forming, the enthalpy released is equal in magnitude to the energy absorbed when the bond breaks (and so we flip the sign).

Bond enthalpies are averages

Looking at C–H bonds in organic molecules, there are many different possible combinations for the structures that can exist. Some organic molecules contain only carbon and hydrogen atoms and others contain electronegative atoms such as oxygen and nitrogen. Because the environment that the C–H bond finds itself in is different in different molecules, the bond energy will also be different. For this reason, bond enthalpies are reported as an average bond enthalpy defined as enthalpy change when one mole of bonds are broken in the gaseous state averaged for the same bond in similar compounds.

Study skills

Bond enthalpies or bond dissociation energies can be found in section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet. Note that these values are averages in units of kJ mol^{-1}.

For example, the O–H bond in water has a slightly different bond enthalpy value compared to the O–H bond in ethanol. Identical bonds in molecules with two (or more) types of bonds also have different bond enthalpy values. In water, for example, it takes more energy to break the first O–H bond than to break the second:

$$H_2O(l) \rightarrow H(g) + OH(g) \quad \Delta H = +502 \text{ kJ mol}^{-1}$$

$$OH(g) \rightarrow H(g) + O(g) \quad \Delta H = +427 \text{ kJ mol}^{-1}$$

To calculate the average bond enthalpy for the two OH bonds in water, we take the average:

$$\begin{aligned}\text{total energy to break both bonds} &= +502 + 427 \\ &= +929 \text{ kJ mol}^{-1}\end{aligned}$$

$$\begin{aligned}\text{average bond enthalpy} &= \frac{+929}{2} \\ &= +464.5 \text{ kJ mol}^{-1}\end{aligned}$$

Compare this with the value in section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet, which is +463 kJ mol^{-1}. It differs slightly because the values in the data booklet are calculated from a range of different compounds, in which the same type of bond has different bond enthalpy values.

In this definition of average bond enthalpy, the word gaseous must be emphasised. Each value that we use from section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet is for gaseous chemical species. If the molecule is in the liquid or solid phase, then even more energy will be required to break the bond since the molecule must first phase change to a gas by overcoming any forces of attraction between the particles themselves.

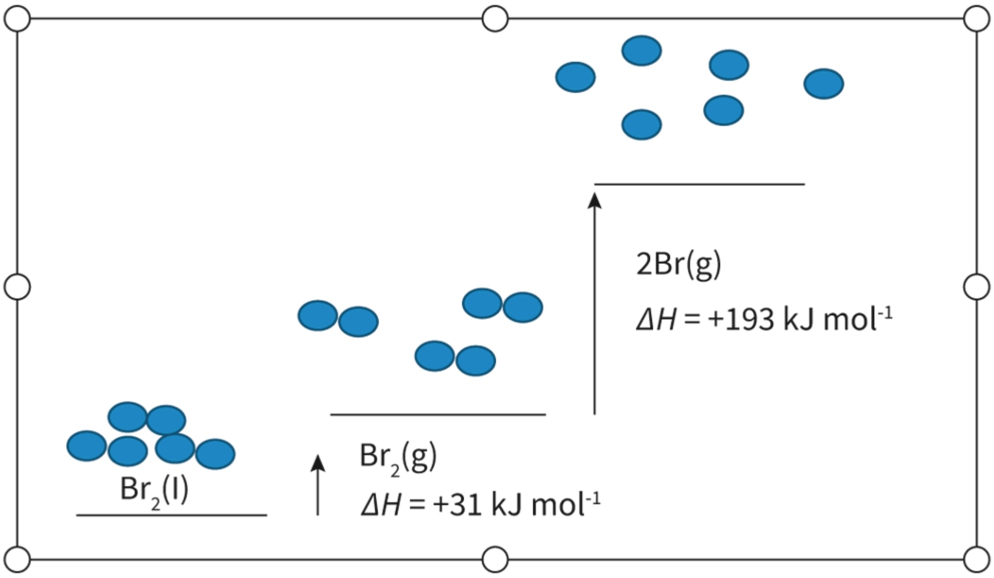


Figure 3. Energy is required to change bromine from a liquid to a gas before the bonds are broken in the gas state.

More information for figure 3

The image is a diagram illustrating the energy changes required to transform bromine from liquid to gas and then to individual gaseous atoms. It shows three stages of bromine (B): liquid (l), gaseous (g), and as separate bromine atoms in gas (g). At the bottom, there are colorful representations of Br(l) molecules. An upward arrow indicates a phase change to Br(g) with an enthalpy change (ΔH) of +31 kJ mol. Further above, another arrow shows the breakup of Br(g) into two bromine atoms, with a larger ΔH value of +193 kJ mol, indicating the energy required to break the bonds in the gas state. The diagram visually represents the stepwise energy increments in the transition from the liquid to the gaseous phase, ending in the dissociation of gaseous molecules into individual atoms.

[Generated by AI]

Calculate the enthalpy of a reaction using bond enthalpies

Average bond enthalpies can be used to calculate the enthalpy change of reactions. Relating back to The big picture, energy is not lost but instead changes forms. In a chemical reaction, energy is absorbed from the surroundings allowing bonds to break. Some of this energy is released again when new bonds are formed and the overall difference in energy allows us to determine whether the reaction is exothermic or endothermic. As such, we can calculate the enthalpy of a reaction by finding the difference in bond energy between the bonds broken and the bonds formed:

$$\Delta H = \Sigma(\text{bond enthalpy (BE) of bonds broken}) - \Sigma(\text{BE of bonds formed})$$

This equation can also be considered as follows:

$$\Delta H = \Sigma(\text{reactant bond enthalpies}) - \Sigma(\text{product bond enthalpies})$$

Nature of Science

Aspect: Models

Scientific models can come in a variety of forms: physical representations, abstract diagrams, mathematical equations or algorithms.

Using average bond enthalpies in the determination of enthalpy of reaction has its limitations. Since bond enthalpies are averages and are derived from a variety of molecules instead of the actual molecules in the reaction you are studying, the resulting answer can be viewed as a highly informed hypothesis as to the actual enthalpy of the reaction. If the reaction were performed in the laboratory, you would likely find that the enthalpy is very close to that which was calculated using the bond enthalpy method. It is for this reason, that modelling the enthalpy of a reaction using the bond enthalpy method of calculation is acceptable.

When determining the enthalpy of a reaction using the bond enthalpy method, it is very important that you correctly balance the reaction and determining the structural formula of each molecule in the reaction. For example, in the reaction:

$$H_2(g) + F_2(g) \rightarrow 2HF(g)$$

two molecules of HF are formed so the bond enthalpy of the H–F bond must be multiplied by two.

Bonds broken:

$$\text{H}-\text{H} \quad 436\ \text{kJ mol}^{-1}$$

$$\text{F}-\text{F} \quad 159\ \text{kJ mol}^{-1}$$

Bonds formed:

$$2\times\text{H}-\text{F} \quad 2\times567\ \text{kJ mol}^{-1}$$

$$\begin{aligned}\Delta H &= \Sigma(\text{BE bonds broken}) - \Sigma(\text{BE bonds formed})\\ &= (436+159)-(2\times567)\\ &= 595-1134\\ &= -539\ \text{kJ mol}^{-1}\end{aligned}$$

Since the enthalpy change for this reaction is negative, it is exothermic.

When using this method, it is important to note that since average bond enthalpies are used there is some inaccuracy. While the order of magnitude of the enthalpy of reaction can be determined, experimental measurements will provide more accurate results.

In what scenarios would the bond enthalpy method be appropriate?

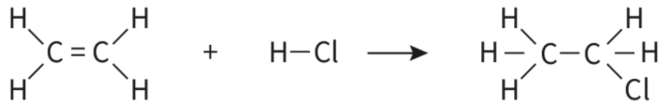
Worked example 1

For the reaction:

$$CH_2{=}CH_2 + HCl \rightarrow CH_3CH_2Cl$$

determine the bonds that are breaking and the bonds that are forming.

To solve questions of this type, the structural formulas of the molecules must first be determined:



Therefore, on the reactants side, the C=C bond and the H−Cl bonds are broken and on the products side, a C−H, C−C and C−Cl are formed.

Worked example 2

Calculate the enthalpy of reaction for the previous reaction using data from section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet.

Bonds broken:

1 × C=C   1 × 614 kJ mol^{-1}

4 × C−H   4 × 414 kJ mol^{-1}

1 × H−Cl   1 × 431 kJ mol^{-1}

Bonds formed:

5 × C−H   5 × 414 kJ mol^{-1}

1 × C−C   1 × 346 kJ mol^{-1}

1 × C−Cl   1 × 324 kJ mol^{-1}

Hence:

$$\begin{aligned}\Delta H &= \Sigma(\text{BE bonds broken}) - \Sigma(\text{BE bonds formed})\\ &= (614 + 431 + 4 \times 414) - (5 \times 414 + 346 + 324)\\ &= 2701 - 2740\\ &= -39 \text{ kJ mol}^{-1}\end{aligned}$$

Kog

Worked example 3

Calculate the enthalpy of reaction:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$$

using data from section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet.

Bonds broken:

$2 \times$ H–H 2×436 kJ mol^{-1}

O=O 498 kJ mol^{-1}

In this example it is important to note the coefficients applied to balance the equation. Since there are two hydrogen molecules, the bond enthalpy for the H–H bond must be multiplied by two.

Bonds formed:

$4 \times$ O–H 4×463 kJ mol^{-1}

As was the case for the two hydrogen molecules, the coefficient in front of the water must also be considered. Since each water molecule contains two O–H bonds and the equation is balanced with a 2 in front of the water, there are four O–H bonds formed.

Next, these values are entered into the equation:

$$\begin{aligned}\Delta H &= \Sigma(\text{BE bonds broken}) - \Sigma(\text{BE bonds formed}) \\ &= (2\times436 + 498) - (4\times463) \\ &= 1370 - 1852 \\ &= -482 \text{ kJ mol}^{-1}\end{aligned}$$

Activity

Kog

- **IB learner profile attribute:** Open-minded
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly.
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair activity

Use common objects to show the bonds that are breaking and forming in a chemical reaction to aid in solving for the enthalpy of reaction.

- Choose a simple chemical reaction and build models of each of the molecules in the reaction. Suitable materials could include toothpicks with candies/dried fruit/pieces of clay, etc. Actual molecular modelling kits could also be used.
- Create a suitable legend for the objects that you have chosen to represent each atom in the reaction.
- Take a photo of the reaction that you have modelled and use that to calculate the enthalpy of the reaction.

R1.2 Energy cycles in reactions

Hess's law

Learning outcomes

By the end of this section you should be able to apply Hess's law to calculate enthalpy changes in multistep reactions.

Imagine you are going for a hike to the top of a mountain and have a choice of paths to take you there. You could go straight up the mountain or take a side route that brings you to a lake or maybe a viewpoint. Whichever path you choose, you end up in the same place (**Figure 1**).

Figure 1. Many paths up a mountain all leading to the same place.

Credit: TravelCouples, Getty Images

Like hiking to the top of a mountain, the net enthalpy of a chemical reaction is independent of the path taken, therefore, enthalpy is a state function. A state function is defined as a property whose value does not depend on the path taken to reach that value. Other examples of state functions that you will have come across in the DP chemistry course are temperature, pressure and volume.

Nature of Science

Aspect: Experiments

Scientists design and perform experiments to obtain data, which can be used to test hypotheses.

Hess's law calculates the enthalpy change for a reaction indirectly through other reactions, rather than directly using experimental evidence for the reaction itself. When might this be a benefit?

Hess’s law, published by Germain Henri Hess in 1840, states that the total enthalpy change in a chemical reaction is independent of the route by which the chemical reaction takes place, as long as the initial and final conditions are the same. This means that we can calculate the enthalpy change of a chemical reaction by considering different possible steps to reach the same products, much like the different pathways that reach the top of the mountain. Hess’s law is an application of the law of conservation of energy since both state that energy is conserved.

Concept

Hess’s law states that the total enthalpy change in a chemical reaction is independent of the route by which the chemical reaction takes place, as long as the initial and final conditions are the same.

Energy cycles

Energy cycles provide us with a useful tool when applying Hess’s law in determining the enthalpy of a reaction. **Figure 2** shows two separate pathways from reactants to products. Both achieve the same product so both pathways could be used to determine the overall enthalpy change for the reaction.

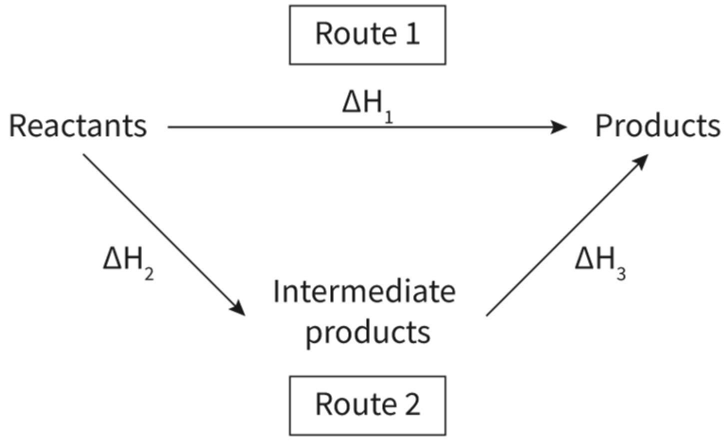


Figure 2. Energy cycle showing two separate pathways to reach the products.

More information for figure 2

The overall enthalpy of the reaction in **Figure 2** could be determined in two ways:

1. Enthalpy of the direct route (**Route 1**) = ΔH_1
2. Enthalpy of the indirect route (**Route 2**) = $\Delta H_2 + \Delta H_3$

From this we can conclude that $\Delta H_1 = \Delta H_2 + \Delta H_3$

Importantly, you can solve for any of the enthalpies in the equation by rearranging the terms:

$$\Delta H_2 = \Delta H_1 - \Delta H_3 \quad \text{and} \quad \Delta H_3 = -\Delta H_2 + \Delta H_1$$

Note that if you go against an arrow in an energy cycle, the value of the enthalpy change is reversed.

Equation method for Hess's law

Calculating the enthalpy of reaction using Hess's law involves manipulating the different pathway equations and cancelling out those atoms or molecules that occur on both sides of the reaction arrow. This method can be thought of as proof that energy cycles work.

Consider the following example for the combustion of pentane:

$$C_5H_{12}(l) + 8O_2(g) \rightarrow 5CO_2(g) + 6H_2O(l)$$

Using the equations below, together with their known enthalpy changes, the enthalpy change of the above reaction can be calculated.

$$5C(s) + 6H_2(g) \rightarrow C_5H_{12}(l) \quad \Delta H = -173 \text{ kJ}$$

$$5C(s) + 5O_2(g) \rightarrow 5CO_2(g) \quad \Delta H = -1970 \text{ kJ}$$

$$6H_2(g) + 3O_2(g) \rightarrow 6H_2O(l) \quad \Delta H = -1716 \text{ kJ}$$

Step 1: Think of each molecule in the reaction you are solving for as a ‘puzzle piece’ where you need to move it to the correct spot:

Reaction you are solving for:

$$\mathbf{C_5H_{12}(l)} + \mathbf{8O_2(g)} \rightarrow \mathbf{5CO_2(g)} + \mathbf{6H_2O(l)}$$

‘Puzzle pieces’:

$$5C(s) + 6H_2(g) \rightarrow \mathbf{C_5H_{12}(l)} \quad \Delta H = -173 \text{ kJ}$$

$$5C(s) + \mathbf{5O_2(g)} \rightarrow \mathbf{5CO_2(g)} \quad \Delta H = -1970 \text{ kJ}$$

$$6H_2(g) + \mathbf{3O_2(g)} \rightarrow \mathbf{6H_2O(l)} \quad \Delta H = -1716 \text{ kJ}$$

From this, we can see that in our ‘puzzle pieces’, $C_5H_{12}(l)$ is on the wrong side of the reaction arrow, so we reverse the reaction and do the same to the enthalpy change:

$$\mathbf{C_5H_{12}(l)} \rightarrow 5C(s) + 6H_2(g) \quad \Delta H = +173 \text{ kJ}$$

In the second and third equations, the O_2, CO_2 and H_2O are on the correct side of the arrow and the sum of the coefficients equals the original reaction equation.

Step 2: Cross off all molecules that appear on both sides of the reaction arrow. The 5C(s) and $6H_2(g)$ on each side of the arrow cancel out.

$$\mathbf{C_5H_{12}(l)} \rightarrow \cancel{5C(s) + 6H_2(g)} \quad \Delta H = +173 \text{ kJ}$$

$$\cancel{5C(s)} + \mathbf{5O_2(g)} \rightarrow \mathbf{5CO_2(g)} \quad \Delta H = -1970\ \text{kJ}$$

$$\cancel{6H_2(g)} + \mathbf{3O_2(g)} \rightarrow \mathbf{6H_2O(l)} \quad \Delta H = -1716\ \text{kJ}$$

Step 3: Write the remaining molecules as a chemical equation. If you have set up your equations properly, you will be left with the reaction that you were solving for.

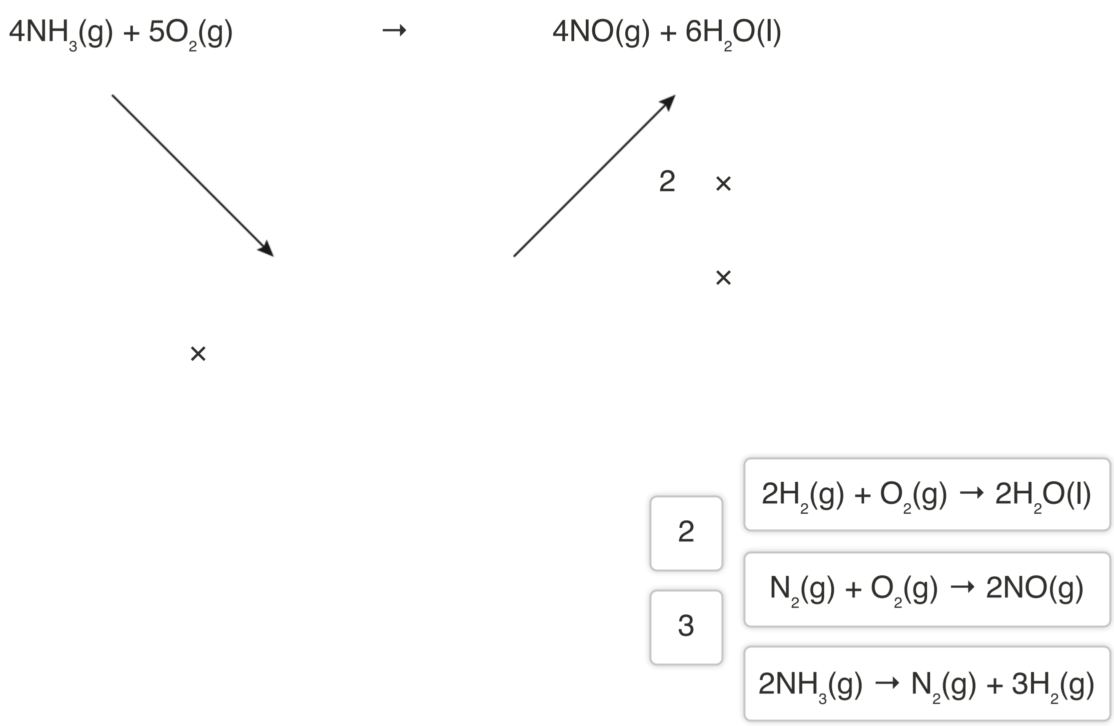
$$C_5H_{12}(l) + 8O_2(g) \rightarrow 5CO_2(g) + 6H_2O(l)$$

Step 4: Calculate the ΔH for the reaction by adding the enthalpy changes for each individual reaction:

$$\Delta H = +173\ \text{kJ mol}^{-1} + (-1970\ \text{kJ mol}^{-1}) + (-1716\ \text{kJ mol}^{-1}) = -3513\ \text{kJ mol}^{-1}$$

Try the example in **Interactive 1** by dragging and dropping the equations and coefficients so the alternative route is equal to the given reaction.

Kog



Check

Interactive 1. Hess’s law.

Study skills

It is very easy to make errors in Hess’s law questions if you are not paying close attention to the states of the chemical species. When you are cancelling out chemical species on either side of the reaction arrow, you can only cancel out those that are in the same state. For example, you can cancel out H_2O(l) with H_2O(l), but **not** H_2O(g) with H_2O(l).

Worked example 1

Using Hess’s law and the provided equations, calculate the enthalpy of reaction for the hydrogenation of propene using Pd as a catalyst:

$$CH_2{=}CHCH_3(g) + H_2(g) \rightarrow CH_3CH_2CH_3(g) \quad \Delta H = ?$$

$$3C(s) + 3H_2(g) \rightarrow CH_2{=}CHCH_3(g) \quad \Delta H_1 = +20\ kJ$$

$$3C(s) + 4H_2(g) \rightarrow CH_3CH_2CH_3(g) \quad \Delta H_2 = -105\ kJ$$

$$\mathbf{CH_2{=}CHCH_3(g)} \rightarrow \cancel{3C(s) + 3H_2(g)} \quad \Delta H_1 = -20\ kJ$$

$$\cancel{3C(s)} + \cancel{4}\ \mathbf{H_2(g)} \rightarrow \mathbf{CH_3CH_2CH_3(g)} \quad \Delta H_2 = -105\ kJ$$

$$CH_2{=}CHCH_3(g) + H_2(g) \rightarrow CH_3CH_2CH_3(g)$$

Therefore,

$$\begin{aligned}\Delta H &= (-1)\Delta H_1 + \Delta H_2 \\ &= (-1)(+20) + (-105) \\ &= -125\ \text{kJ mol}^{-1}\end{aligned}$$

Worked example 2

Using the data provided, what is the enthalpy of reaction for the following?

$$5C(s) + 6H_2(g) \rightarrow CH_3CH_2CH_2CH_2CH_3(l) \quad \Delta H = ?$$

$$C(s) + O_2(g) \rightarrow CO_2(g) \quad \Delta H_1 = -394\ kJ$$

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l) \quad \Delta H_2 = -286\ kJ$$

$$CH_3CH_2CH_2CH_2CH_3(l) + 8O_2(g) \rightarrow 5CO_2(g) + 6H_2O(l) \quad \Delta H_3 = -3509\ kJ$$

$$\mathbf{5C(s)} + \cancel{5O_2(g) \rightarrow 5CO_2(g)} \quad \Delta H_1 = -394\ kJ \ \times 5$$

$$\mathbf{6H_2(g)} + \cancel{3O_2(g)} \rightarrow \cancel{6H_2O(l)} \quad \Delta H_2 = -286\ kJ \ \times 6$$

$\cancel{5CO_2(g)}$ + $\cancel{6H_2O(l)}$ ⇢ $\mathbf{CH_3CH_2CH_2CH_2CH_3(l)}$ + $\cancel{8O_2(g)}$
ΔH_3 = 3509 kJ × 1

5C(s) + 6H_2(g) ⇢ $CH_3CH_2CH_2CH_2CH_3$(l)

Therefore,

$$\begin{aligned}\Delta H &= 5\Delta H_1 + 6\Delta H_2 - \Delta H_3\\ &= 5(-394) + 6(-286) + (3509)\\ &= -177\ \text{kJ mol}^{-1}\end{aligned}$$

Worked example 3

4C(s) + 6H_2(g) + O_2(g) ⇢ 2CH_3CH_2OH(l) ΔH = ?

C(s) + O_2(g) ⇢ CO_2(g) ΔH_1 = –394 kJ

H_2(g) + ½O_2(g) ⇢ H_2O(l) ΔH_2 = –286 kJ

CH_3CH_2OH(l) + 3O_2(g) ⇢ 2CO_2(g) + 3H_2O(l) ΔH_3 = –1367 kJ mol^{-1}

4C(s) + $\cancel{4O_2(g) \rightarrow 4CO_2(g)}$ ΔH_1 = –394 kJ × 4

$\mathbf{6H_2(g)}$ + $\cancel{3}$ $\mathbf{O_2(g)}$ ⇢ $\cancel{6H_2O(l)}$ ΔH_2 = –286 kJ × 6

$\cancel{4CO_2(g) + 6H_2O(l)}$ ⇢ $\mathbf{2CH_3CH_2OH(l)}$ + $\cancel{6O_2(g)}$
ΔH_3 = –1367 kJ × –2

4C(s) + 6H_2(g) + O_2(g) ⇢ 2CH_3CH_2OH(l)

Therefore,

$$\begin{aligned}\mathit{\Delta} H &= 4\mathit{\Delta} H_1 + 6\mathit{\Delta} H_2 + (-2)\mathit{\Delta} H_3\\ &= (4)(-394) + (6)(-286) - (2)(-1367)\\ &= -558\ \text{kJ mol}^{-1}\end{aligned}$$

Alternate method for Hess’s law

Hess’s law can also be approached using an energy cycle diagram method. Mathematically, the answer will be the same as the equation method previously presented, but this method provides an illustrative presentation of the equations instead.

Study skills

There is always more than one way to solve a question and your job as a learner is to determine what works best for you. The equation method for answering Hess’s law question is a perfect example of this! Keep in mind that you should be able to answer questions using both methods, but when given a choice it is up to you.

$$C(graphite) + 2H_2(g) \rightarrow CH_4(g)$$

Instead of using chemical equations for the separate pathways, a diagram can be used instead. In the diagram shown in **Figure 3**, the equations provided are for the combustions of each chemical species in the equation that you are solving for (the formation of methane).

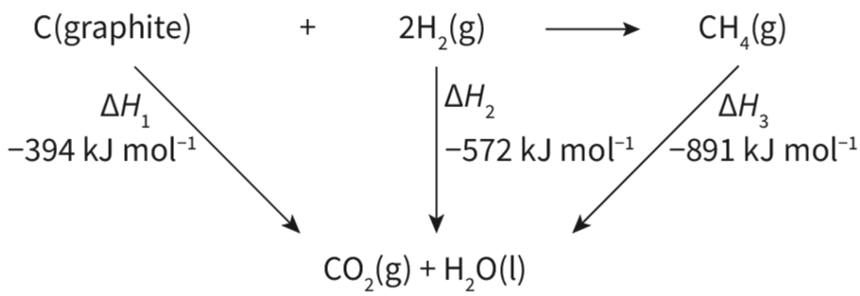


Figure 3: Energy cycle for Hess’s law for the formation of methane from its elements.

More information for figure 3

In this reaction, we can see that two moles of hydrogen gas are required to form one mole of methane. Therefore, we must multiply the enthalpy for hydrogen gas by two:

$$\begin{aligned} \Delta H_2 &= 2\times -286 \text{ kJ mol}^{-1} \\ &= -572 \text{ kJ mol}^{-1} \end{aligned}$$

Kog

To determine the enthalpy of the reaction we will use:

$$\Delta H_r = \Delta H_1 + \Delta H_2 - \Delta H_3$$

We need to imagine that we are ‘flipping the arrow’ for ΔH_3 so that it is pointed at the products. For this reason, we need to change the sign to a negative. This was the same strategy that we used for the equation method when we multiplied by a negative number if we need to flip the reaction from reactants to products, to products to reactants instead. If we were to write out the equations, we would see that this allows us to properly cancel out all the chemical species which are not in the overall equation.

Therefore:

$$\begin{aligned}\Delta H_r &= [(-394) + (2\times - 286)] - (-891)] \\ &= -75 \text{ kJ mol}^{-1}\end{aligned}$$

Sometimes one of the hardest parts of solving Hess’s law questions is setting up the equations. Use **Interactive 2** to arrange the energy cycle to determine the enthalpy of reaction for:

$$4C(s) + 5H_2(g) \rightarrow C_4H_{10}(g)$$

$4C(s) + 5H_2(g) \rightarrow C_4H_{10}(g)$

×
–394
KJ mol^{-1}

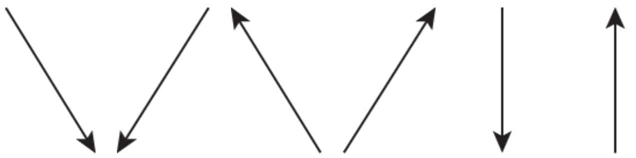
×
–286
KJ mol^{-1}

×
–2878
KJ mol^{-1}

carbon dioxide + water

ΔH = KJ mol^{-1}

		32
1	–1	–32
2	–2	64
3	–3	–64
4	–4	128
5	–5	–128



Check

Interactive 2. Enthalpy of reaction of carbon and hydrogen to give butane.

Creativity, activity, service

Strand: Service
Learning outcome: Demonstrate engagement with issues of global significance

Heat is a form of energy and is defined as the movement of energy from a hotter temperature object to a cooler one. We often feel heat changes in our daily lives but how do changes in heat affect the bigger picture? One such example is global warming and climate change (https://climate.nasa.gov/global-warming-vs-climate-change/) which could have devastating effects on the environment.

Kog

As an IB student, think about how you can demonstrate engagement with issues of global significance such as these. Remember that small changes made in your local community can lead to big changes on a global scale. 'Think globally, act locally' is a phrase that you should keep in mind when carrying out your CAS experiences.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learnin**g: Thinking skills— Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual/pair activity

How do the states/phases of chemical species impact the enthalpy of reaction and how can we use Hess's law to investigate this?

- Using Hess's law, solve for the enthalpy of reaction for the following:
    - Solid NaOH dissolved in water
    - Aqueous NaOH reacted with aqueous HCl
    - Solid NaOH reacted with aqueous HCl
- Compare and contrast the enthalpies and discuss how they relate to one another.
- Write a conclusion statement that explains the impact of the state of a substance on the overall enthalpy of a reaction with reference to the NaOH/HCl chemical system.

R1.2 Energy cycles in reactions

Standard enthalpy changes of combustion and formation (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R1.2.3) Deduce equations and solutions to problems involving enthalpy changes of combustion and formation.
- (R1.2.4) Calculate enthalpy changes of a reaction using $\Delta H_f^{\ominus}$ data or $\Delta H_c^{\ominus}$ data:

$$\Delta H^{\ominus} = \Sigma(\Delta H_f^{\ominus} \text{ products}) - \Sigma(\Delta H_f^{\ominus} \text{ reactants})$$
$$\Delta H^{\ominus} = \Sigma(\Delta H_c^{\ominus} \text{ reactants}) - \Sigma(\Delta H_c^{\ominus} \text{ products})$$

For every chemical process and many physical changes (such as change of state) there is an associated enthalpy value. For example, there are enthalpies of neutralisation, enthalpies of combustion and enthalpies of vaporisation (for liquids transitioning to the gaseous state). Enthalpies of reaction are very important for chemists as a way to understand chemical stability and the tendency for the reaction to occur. Two such important examples are standard enthalpy changes of combustion, $\Delta H_c^{\ominus}$, and formation, $\Delta H_f^{\ominus}$. For these enthalpies, the values are determined under very specific conditions using apparatus similar to that shown in **Figure 1**.

Why do you think that it is so important to control the conditions when determining the standard enthalpies of formation and combustion?

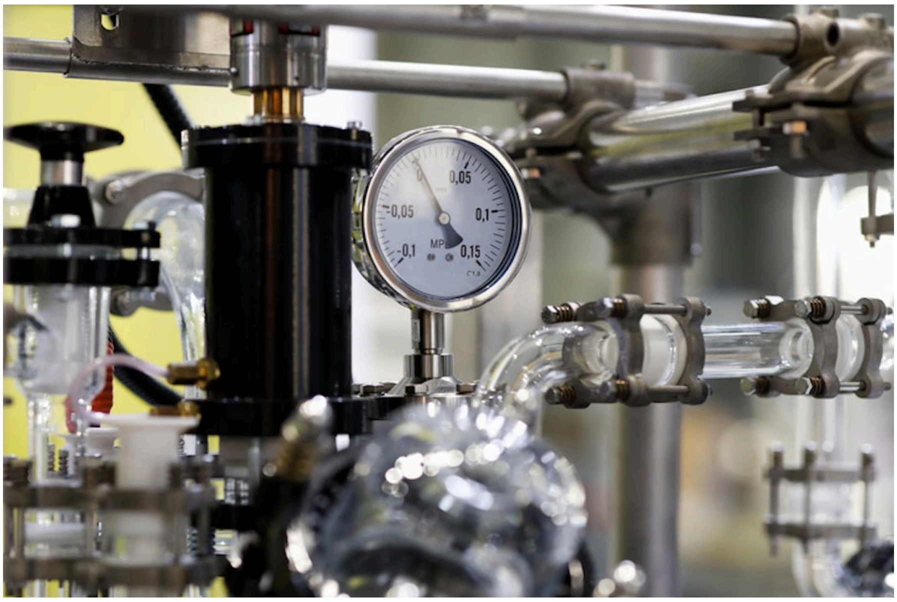


Figure 1: Regulators used to control the atmospheric conditions of an experiment.

Credit: genkur, Getty Images

Standard enthalpy changes of combustion, $\Delta H_c^{\ominus}$

Combustion can be defined as an exothermic chemical process where a substance is combined with oxygen gas. Hydrocarbon combustion is a common example for combustion reactions, but metals can combust, too! For example, magnesium metal combusts to form magnesium oxide:

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

You will learn much more about combustion in section R1.3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/combustion-reactions-id-43476/). In this section we will focus on using standard enthalpies of combustion in Hess's law questions.

The molar enthalpy of combustion, or the standard enthalpy of combustion ($\Delta H_c^{\ominus}$), for a substance is defined as the enthalpy change when one mole of a substance is burned completely in oxygen under standard state conditions. The thermochemical equation for the combustion of ethanol is shown below:

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \quad \Delta H_c^{\ominus} = -1367 \text{ kJ mol}^{-1}$$

Understanding, measuring, and comparing standard enthalpies of combustion are very useful in evaluating the efficiency of fuels and determining the calories in food.

Nature of Science

Aspect: Science as a shared endeavour

Climate change with respect to combustion of hydrocarbons

Science drives a portion of governmental policy, and when communicated effectively it has the power to shift the attitude and behaviour of societies. We have seen this recently with societal shifts from the COVID pandemic and climate change. Since both the pandemic and climate change are worldwide issues, it highlights the importance of collaboration. How can we support and improve collaboration worldwide to support progress toward shared goals?

Standard enthalpy of combustion values are found in section 14 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/) of DP chemistry data booklet and refer to values determined at a temperature of 298.15 K and a pressure of 100 kPa.

Another example of a thermochemical equation is shown. This equation is for the complete combustion of one mole of propane, $CH_3CH_2CH_3$.

$$CH_3CH_2CH_3(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l) \quad \Delta H_c^{\ominus} = -2219 \text{ kJ mol}^{-1}$$

Study skills

It is important to note that the values that you find in section 14 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/) of the DP chemistry data booklet are for one mole of the substance that is combusted. Sometimes this requires fractions when balancing the equation. In order to arrive at integer coefficients, it is important to remember that you need to multiply the enthalpy value by the same number that you used to multiply the coefficients.

Consider this example for the combustion of hydrogen gas:

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \quad \Delta H_c = -242 \text{ kJ mol}^{-1}$$

Note that $\frac{1}{2}O_2$ was required to balance this equation so that there was one mol of H_2O. In order to arrive at only integers for the coefficients, the entire equation needs to be multiplied by two — including the enthalpy value:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$
$$\Delta H_c = -242 \text{ kJ mol}^{-1} \times 2 = -484 \text{ kJ mol}^{-1}$$

Worked example 1

Write the balanced equation for the combustion for ethene (CH_2CH_2).

$$CH_2CH_2(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(l)$$

Worked example 2

Write the balanced equation for the combustion for benzoic acid (C_6H_5COOH).

$$C_6H_5COOH(g) + 7\frac{1}{2}O_2(g) \rightarrow 7CO_2(g) + 3H_2O(l)$$

Standard enthalpy changes of formation, $\Delta H_f^{\ominus}$

The molar enthalpy of formation or the standard enthalpy of formation ($\Delta H_f^{\ominus}$) for a substance is defined as the energy change that occurs when one mole of a substance is formed from its constituent elements in their standard state.

The constituent elements of a compound must be included in the state at which they exist under standard conditions. Most elements are solids under standard conditions with the exception of eleven elements which are gases and two which are liquids (Hg and Br_2). Importantly, some of these gases and liquids must always be written in their diatomic form (H_2, Cl_2, Br_2, etc).

Making connections

Allotropes of carbon (diamond, graphite, fullerenes, and graphene) are all technically carbon in its standard state (see section S2.2.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-

753301/book/covalent-network-structures-id-43593/)). Would you expect allotropes of an element, such as diamond and graphite, to have different $\Delta H_f^{\ominus}$ values?

Standard enthalpy of formation values are found in section 13 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) of the DP chemistry data booklet. Like standard enthalpies of combustion, this information can be conveyed in an equation together with the change in enthalpy.

$$C(s) + \frac{3}{2}H_2(g) + \frac{1}{2}Cl_2(g) \rightarrow CH_3Cl(g) \quad \Delta H_f^{\ominus} = -82 \text{ kJ mol}^{-1}$$

Study skills

As was the case for standard enthalpies of combustions, the values in section 13 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) of the DP chemistry data booklet are for the formation of one mol of the product. Sometimes this requires fractions to be used in the balancing.

It is also important to note that all reactants must be written as elements in their standard state. Typically this will be a single atom, except for diatomic elements:

H_2, F_2, Cl_2, Br_2, I_2, N_2, O_2

Worked example 3

Write the balanced equation for the formation of benzene (C_6H_6) from its elements.

$$6C(s) + 3H_2(g) \rightarrow C_6H_6(l)$$

Worked example 4

Write the balanced equation for the formation of HF, stating $\Delta H_f^{\ominus}$ as found in the DP chemistry data booklet in section 13 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/).

$$\frac{1}{2}H_2(g) + \frac{1}{2}F_2(g) \rightarrow HF(g) \quad \Delta H_f{}^{\theta} = -273 \text{ kJ mol}^{-1}$$

Study skills

Be really careful when choosing values from section 13 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) of the DP chemistry data booklet as the magnitude of the enthalpy of formation is dependent on the state of the molecule.

For example, the standard enthalpy of formation for liquid water is −286 kJ mol^{-1} while the standard enthalpy of formation for gaseous water (steam) is −242 kJ mol^{-1}.

How can you explain the difference in magnitude for the enthalpies of formation for liquid versus gaseous water?

Hess's law and standard enthalpies of combustion

Calculating standard enthalpy changes of combustion, $\Delta H_c^\ominus$, and formation, $\Delta H_f^\ominus$ is an application of Hess's law. Therefore, standard enthalpies of combustion or formation can be used to calculate the enthalpy of reaction using an energy cycle like what we saw in section R1.2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/hesss-law-id-46252/).

For combustion, we can look at this using the following equation where the enthalpy of reaction is the difference between the sum of reactant enthalpies of combustion and the sum of the product enthalpies of combustion. Importantly, each enthalpy must be multiplied by the stoichiometric balancing term (coefficient) in the balanced equation before the enthalpies are summed.

$$\Delta H^\ominus = \Sigma(\Delta H_c^\ominus \text{ reactants}) - \Sigma(\Delta H_c^\ominus \text{ products})$$

Below is an example for determining the enthalpy for the formation of glucose using the standard enthalpies of combustion (from section 14 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/) of the DP chemistry data booklet) of glucose and its constituent elements. This example is presented in **Table 1** to assist with organising the data.

Table 1. Determining the enthalpy for the formation of glucose.

	$6C(s)$	+	$6H_2(g)$	+	$3O_2(g)$	→	C_6H_{12}
	$\Delta H^\ominus = ?$						
$\Delta H_c^\ominus$ (kJ mol^{-1})	−394		−286		0		−28
Coefficient	6		6		3		1
Total enthalpy (kJ mol^{-1})	= (−394) × 6 = −2364		= (−286) × 6 = −1716		= (0) × 3 = 0		= (−28C = −280.
Sums (kJ mol^{-1})	$\Sigma(\Delta H_c^\ominus$ reactants) = (−2364) + (−1716) +(0) = −4080						$\Sigma(\Delta H_c^\ominus$ product = (−28C = −280.

$$\begin{aligned}\Delta H^\ominus &= \Sigma(\Delta H_c^\ominus \text{ reactants}) - \Sigma(\Delta H_c^\ominus \text{ products})\\ &= (-4080) - (-2803)\\ &= -1277 \text{ kJ mol}^{-1}\end{aligned}$$

Worked example 5

Using the standard enthalpies of combustion from section 14 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/) of the DP chemistry data booklet, calculate the enthalpy of reaction for the desaturation of cyclohexane to benzene:

$$C_6H_{12}(l) \rightarrow C_6H_6(l) + 3H_2(g)$$

	$C_6H_{12}(l)$	→	C_6
$\Delta H_c^{\ominus}$ (kJ mol^{-1})	−3920		−3
Coefficient	1		
Total enthalpy (kJ mol^{-1})	−3920		−3
Sums (kJ mol^{-1})	$\Sigma(\Delta H_c^{\ominus}$ reactants) = −3920		$\Sigma(\Delta H$ = (−3 = −41

$$\begin{aligned}\Delta H^{\ominus} &= \Sigma(\Delta H_c^{\ominus} \text{ reactants}) - \Sigma(\Delta H_c^{\ominus} \text{ products}) \\ &= (-3920) - (-4126) \\ &= 206 \text{ kJ mol}^{-1}\end{aligned}$$

Worked example 6

Using the standard enthalpies of combustion from section 14 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/) of the DP chemistry data booklet, calculate the enthalpy of reaction for the hydrogenation of buta-1,3-diene:

$$C_4H_6(g) + 2H_2(g) \rightarrow C_4H_{10}(g)$$

	$C_4H_6(g)$	+	2H
$\Delta H_c^{\ominus}$ (kJ mol^{-1})	−2541		−2
Coefficient	1		[illegible]
Total enthalpy (kJ mol^{-1})	−2541		−5

	$C_4H_6(g)$	+	2H
Sums ($kJ\ mol^{-1}$)	$\Sigma(\Delta H_c^\ominus$ reactants) = (−2541) + (−572) = −3113		$\Sigma(\Delta H$ = −2

$$\begin{aligned}\Delta H^\ominus &= \Sigma(\Delta H_c^\ominus \text{ reactants}) - \Sigma(\Delta H_c^\ominus \text{ products})\\ &= (-3113) - (-2878)\\ &= -235 \text{ kJ mol}^{-1}\end{aligned}$$

International Mindedness

An understanding of combustion enthalpies is useful in helping us to select fuels that are more climate conscious. While most of the world's energy supply comes from the combustion of hydrocarbons, scientists are actively optimising alternatives that deliver comparable energy in terms of efficiency, while emitting less harmful by-products. Scientific advances and political focus differ by country, though. How can we, as a global society, work together to mitigate the harmful effects of climate change from the combustion of hydrocarbons?

Hess's law and standard enthalpies of formation

In a nearly identical process, standard enthalpies of formation can be used as an application of Hess's law to determine the enthalpy of a reaction that cannot be measured directly.

$$\Delta H^\ominus = \Sigma(\Delta H_f^\ominus \text{ products}) - \Sigma(\Delta H_f^\ominus \text{ reactants})$$

This equation is the difference between the sum of product enthalpies of formation and the sum of the reactant enthalpies of formation. Like combustion, we must remember to multiply the standard enthalpies of formation by the stoichiometric balancing term (coefficient) in the balance equation before the enthalpies are summed.

Theory of Knowledge

Enthalpy of formation equations represent chemical reactions that do not necessarily take place in the natural world. Is it possible to use these equations as evidence to acquire further knowledge if they cannot be observed?

Consider the following example for determining the enthalpy for the combustion of propane using the standard enthalpies of formation (from section 13 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) of the DP chemistry data booklet) of propane, carbon dioxide, and water. This example is also presented in a **Table 2** to assist with organising the data.

Table 2. Determining the enthalpy for the combustio

	$C_3H_8(g)$	+	$5O_2(g)$	→	3CO
					ΔH° = ?
$\Delta H_f^\ominus$ (kJ mol^{-1})	−105		0		−39
Coefficient	1		5		3
Total enthalpy (kJ mol^{-1})	= (−105) × 1 = −105		= (0) × 5 = 0		= (−39 = −1182
Sums (kJ mol^{-1})	Σ($\Delta H_f^\ominus$ reactants) = (−105) + (0) = −105				Σ($\Delta H_f^\ominus$ = (−118 = −232

$$\begin{aligned}\Delta H^\ominus &= \Sigma(\Delta H_f^\ominus \text{ products}) - \Sigma(\Delta H_f^\ominus \text{ reactants})\\ &= (-2326) - (-105)\\ &= -2221 \text{ kJ mol}^{-1}\end{aligned}$$

Worked example 7

Using the standard enthalpies of formation from section 13 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) of the DP chemistry data booklet, calculate the enthalpy of reaction for the combustion of phenylethene:

$$C_6H_5CHCH_2(l) + 10O_2(g) \rightarrow 8CO_2(g) + 4H_2O(l)$$

	$C_6H_5CHCH_2(l)$	+	$10O_2(g)$	-
$\Delta H_f^\ominus$ (kJ mol^{-1})	104		0	
Coefficient	1		10	
Total enthalpy (kJ mol^{-1})	104		0	
Sums (kJ mol^{-1})	Σ($\Delta H_f^\ominus$ reactants) = 104			

$$\begin{aligned}\Delta H^{\ominus} &= \Sigma(\Delta H_f^{\ominus}\text{ products}) - \Sigma(\Delta H_f^{\ominus}\text{ reactants})\\ &= (-4296) - (104)\\ &= -4400\text{ kJ mol}^{-1}\end{aligned}$$

Worked example 8

Calculate the enthalpy for the reaction of nitrogen dioxide with water using the data provided.

$$2NO_2(g) + 2H_2O(g) \rightarrow 3O_2(g) + N_2H_4(g)$$

$\Delta H_f^{\ominus}$ NO_2 = 33 kJ mol^{-1}

$\Delta H_f^{\ominus}$ $H_2O(g)$ = –242 kJ mol^{-1}

$\Delta H_f^{\ominus}$ N_2H_4 = 48 kJ mol^{-1}

	$2NO_2(g)$	+	$2H_2O(g)$	→
$\Delta H_f^{\ominus}$ (kJ mol^{-1})	33		−242	
Coefficient	2		2	
Total enthalpy (kJ mol^{-1})	66		−484	
Sums (kJ mol^{-1})	$\Sigma(\Delta H_f^{\ominus}$ reactants) = 66 + (−484) = −418			

$$\begin{aligned}\Delta H^{\ominus} &= \Sigma(\Delta H_f^{\ominus}\text{ products}) - \Sigma(\Delta H_f^{\ominus}\text{ reactants})\\ &= (48) - (-418)\\ &= 466\text{ kJ mol}^{-1}\end{aligned}$$

Worked example 9

Calculate the enthalpy for the formation of propan-1-ol using the data provided.

$$3C(s) + 4H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow CH_3CH_2CH_2OH(l) \quad \Delta H = ?$$

$\Delta H_c^{\ominus}(CH_3CH_2CH_2OH)$ = –2021 kJ mol^{-1}

$\Delta H_f^{\ominus}(H_2O(l))$ = –286 kJ mol^{-1}

$\Delta H_f^{\ominus}(CO_2)$ = –394 kJ mol^{-1}

$CH_3CH_2CH_2OH(l) + \frac{9}{2}O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
$\Delta H_c^{\ominus}$ = −2021 kJ mol^{-1} ×1

$4H_2(g) + 2O_2(g) \rightarrow 4H_2O(l)$ $\Delta H_f^{\ominus}$ = −286 kJ mol^{-1} ×4

$3C(s) + 3O_2(g) \rightarrow 3CO_2(g)$ $\Delta H_f^{\ominus}$ = −394 kJ mol^{-1} ×3

Therefore:

$3CO_2(g) + 4H_2O(l) \rightarrow CH_3CH_2CH_2OH(l) + \frac{9}{2}O_2(g)$
$\Delta H_c^{\ominus} = +\ 2021\ \text{kJ mol}^{-1}$

$4H_2(g) + 2O_2(g) \rightarrow 4H_2O(l) \quad \Delta H_f^{\ominus} = -1144\ \text{kJ mol}^{-1}$

$3C(s) + \frac{7}{2}O_2(g) \rightarrow 3CO_2(g) \quad \Delta H_f^{\ominus} = -1182\ \text{kJ mol}^{-1}$

$3C(s) + 4H_2(g) + \frac{1}{2}O_2(g) \rightarrow CH_3CH_2CH_2OH(l)$
$\Delta H = 2021 - 1144 - 1182$
$= -305\ \text{kJ mol}^{-1}$

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 20 minutes
- **Activity type:** Group activity

In groups, work together to solve for the enthalpy of combustion of each of the fuels, listed below.

- Fuels
  - butane
  - propane
  - octane
  - ethanol
  - methane
- For each fuel type, one group member will be responsible for finding the enthalpy of formation for the reactants, another will be responsible for finding the enthalpy of formation of the products, and the group member will perform the calculation to determine the overall enthalpy.
- Make sure to rotate roles in the group for each question.

R1.2 Energy cycles in reactions

Born-Haber cycle (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Interpret and determine values from a Born-Haber cycle for compounds composed of univalent and divalent ions.
- Understand the following terms and use those values in Born-Haber cycle calculations: ionisation energies, enthalpy of atomisation (using sublimation and/or bond enthalpies), electron affinities, lattice enthalpy, enthalpy of formation.

Salts are common compounds found in our bodies, in the environment, and in the laboratory. But how are they formed from their elements? This is not a common reaction in the environment, but under certain conditions, can be completed in a laboratory. The elements would need to undergo a number of transformations in order to be in a cationic or anionic form ready for bonding in a crystal lattice. The metal would begin as a solid that would need to be separated from the other atoms held by metallic bonding and the non-metal would need to be separated from its covalent bonds and intermolecular forces. And once separated, the metal would need to lose electrons with the non-metal gaining electrons. How many steps are involved and what is the sequence?

Figure 1. Great Salt Lake, Utah, USA.

Credit: Alex Potemkin, Getty Images

Making connections

Section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) showed us that the ionic bond is formed by electrostatic attractions between oppositely charged ions and in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/) we saw that ionic compounds exist as three-dimensional lattice structures, represented by empirical formulas. How can we assess the magnitude of lattice enthalpy from the chemical structure?

Processes to transform atoms and molecules into salts

As we learned in sections S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) and S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/) three-dimensional lattice structures, or salts are composed of ions which are bonded ionically. Their formation can be expressed with the enthalpy of formation, $\Delta H_f^{\ominus}$ defined as the enthalpy change that would occur when one mole of a compound is formed from its elements in their standard states under standard conditions.

But it is not as simple as mixing a metal with a non-metal – many steps are required.

To illustrate these steps, let's investigate the formation of NaCl from its elements, Na(s) and Cl_2(g). This overall reaction would be measured with the standard enthalpy of formation:

$$\mathrm{Na(s)} + \tfrac{1}{2}\mathrm{Cl_2(g)} \rightarrow \mathrm{NaCl(s)} \qquad \Delta H_f^{\ominus}\,(\mathrm{NaCl}) = -411\ \mathrm{kJ\ mol^{-1}}$$

Firstly, solid sodium metal would need to be transformed into gaseous sodium which would be calculated with the enthalpy of atomisation.

- Enthalpy of atomisation, $\Delta H^{\ominus}_{\mathrm{atom}}$: the enthalpy change when one mole of gaseous atoms is formed from an element in its standard state.

For metals:

$$\mathrm{M(s)} \rightarrow \mathrm{M(g)} \qquad \Delta H^{\ominus}_{\mathrm{atom}} > 0$$

For diatomic species:

$$\tfrac{1}{2}\mathrm{X_2(g)} \rightarrow \mathrm{X(g)} \qquad \Delta H^{\ominus}_{\mathrm{atom}} > 0$$

Therefore:

$$\mathrm{Na(s)} \rightarrow \mathrm{Na(g)} \qquad \Delta H^{\ominus}_{\mathrm{atom}}\,(\mathrm{Na}) = +110\ \mathrm{kJ\ mol^{-1}}$$

Once the metal is in the gaseous state, electrons need to be removed to create cations. This is measured using the ionisation energy:

- Ionisation energy, IE: the energy required to remove one mole of electrons from one mole of gaseous atoms. For some metals with multiple valence electrons, the second ionisation energy may be required.

IE_1:

$$\mathrm{M(g)} \rightarrow \mathrm{M^+(g)} + e^- \qquad IE_1 > 0$$

IE_2:

$$M^+(g) \rightarrow M^{2+}(g) + e^- \quad IE_2 > 0$$

Therefore:

$$Na(g) \rightarrow Na^+(g) + e^- \quad IE\ (Na) = +496\ kJ\ mol^{-1}$$

Next, we need to think about how to change the non-metal, $Cl_2(g)$ into anions. The first step is breaking the bond between the chlorine atoms. This is measured using bond energy:

- Bond energy, E: the energy required to break one mole of bonds in the gaseous state.

For chlorine, we can write this as follows:

$$Cl_2(g) \rightarrow 2Cl(g) \quad E(Cl_2) = +242\ kJ\ mol^{-1}$$

With the non-metal now existing as an atom instead of a molecule, electrons need to be added to form an anion. This is measured with electron affinity:

- Electron affinity, EA: the energy released when one mole of electrons is added to one mole of gaseous atoms.

$$X(g) + e^- \rightarrow X^-(g) \quad EA < 0$$

Therefore:

$$Cl(g) + e^- \rightarrow Cl^-(g) \quad EA\ (Cl) = -349\ kJ\ mol^{-1}$$

Study skills

Finding the data required when solving questions with lattice enthalpy can be one of the trickiest parts! Here is where to find the data:

- Enthalpy of atomisation, $\Delta H^{\ominus}_{atom}$: will be given in the question or you will be asked to solve for it.
- Ionisation energy, IE : is found in section 9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet.
- Bond energy, *E* : found in section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet.
- Electron affinity, EA : found in section 9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet .
- Standard enthalpy of formation, $\Delta H^{\ominus}_f$: found in section 13 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) of the DP chemistry data booklet .

Born-Haber cycle

Recall from section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/) that lattice enthalpy, $\Delta H^{\ominus}_{lattice}$, gives us information about the strength of ionic bonds in an ionic compound. Lattice enthalpy is the process by which gaseous ions are formed from one mole of solid ionic compound. This process is endothermic and can be represented by the following equation where MX(s) represents one mole of solid ionic compound.

$$MX(s) \rightarrow M^{+}(g) + X^{-}(g) \quad \Delta H^{\ominus}_{lattice} > 0$$

Hess's law provides us with a tool to determine the lattice enthalpy of ionic compounds. This is particularly powerful since it is not possible to determine the lattice enthalpy of an ionic compound directly by experiment.

Study skills

Pay careful attention to the sign of lattice enthalpy. The IB defines lattice enthalpy as the endothermic process of the crystal lattice breaking into its ion:

$$MX(s) \rightarrow M^{+}(g) + X^{-}(g) \quad \Delta H^{\ominus}_{lattice} > 0$$

Values for experimental lattice enthalpies at 298.15 K are found in section 16 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/lattice-enthalpies-at-29815-k-experimental-values-id-45118/) of the DP chemistry data booklet.

The energy cycle that we use for determining the lattice enthalpy of an ionic compound is called a Born-Haber cycle. This energy cycle follows the law of conservation of energy (like all other energy cycles in this section) and it presents a visual representation of the process.

Nature of Science

Aspect: Models

The sign (+ or −) of lattice enthalpy varies, depending on the perspective of the definition: destruction or formation of the crystal lattice. Does it make more sense for the energy holding two ions together to be represented with a positive or negative sign?

Looking back to our previous example with NaCl, let's use the Born-Haber cycle to solve for its lattice enthalpy.

A Born-Haber cycle is structured as follows using the enthalpy of atomisation, ionisation energy, electron affinity, and enthalpy of formation as shown in **Figure 2**.

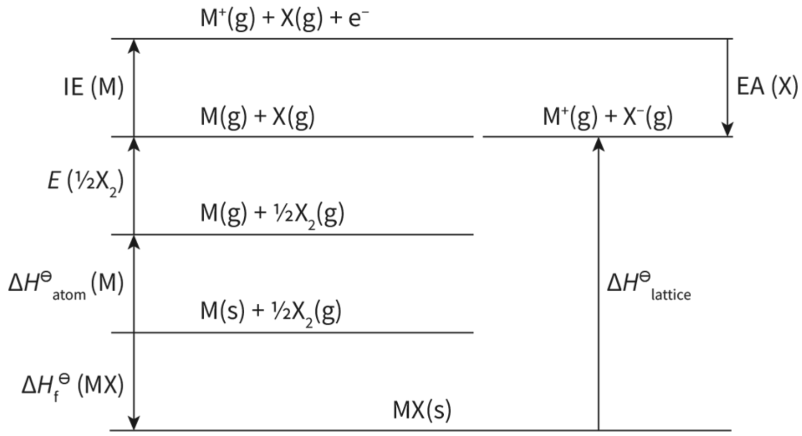


Figure 2. General Born-Haber cycle.

More information for figure 2

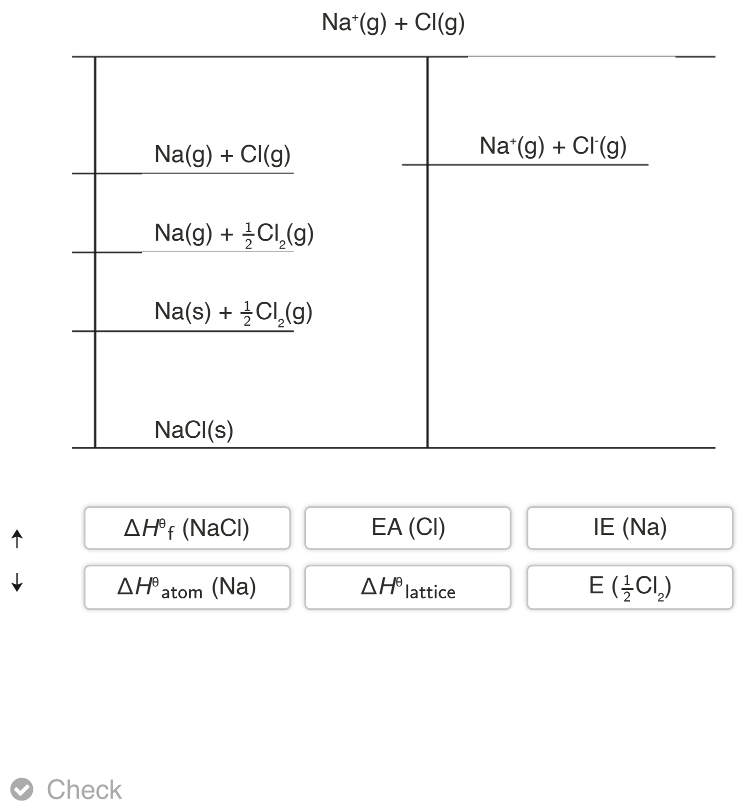
The image is a diagram illustrating a Born-Haber cycle, showing energy changes during the formation of an ionic compound from its elements. The diagram includes several horizontal lines connected by vertical arrows, each representing different energy stages:

1. At the bottom, the equation is (MX(s)), representing the solid state of the compound.
2. Above this, during the enthalpy of formation (ΔH_f^θ(MX)), the conversion from solid to gas is represented as (M(s) + \frac{1}{2}X_2(g)).
3. The next stage is the enthalpy of atomization (ΔH_atom^θ (M)), depicted as (M(g) + \frac{1}{2}X_2(g)).
4. This is followed by the bond dissociation energy (E(\frac{1}{2}X_2)), forming (M(g) + X(g)).
5. Ionization energy (IE(M)) is applied to obtain (M^+(g) + X(g) + e^-).
6. Finally, electron affinity (EA(X)) leads to (M^+(g) + X^-(g)), completing the cycle.

Arrows indicate the direction and type of energy change at each stage, linking the equations sequentially.

[Generated by AI]

Try using **Interactive 1** to create the Born-Haber cycle for NaCl.



Interactive 1. Born-Haber Cycle for NaCl.

More information for interactive 1

An interactive diagram illustrates the Born-Haber cycle for NaCl. Users need to drag and drop the appropriate thermodynamic terms and directional arrows (upward and downward) to complete the cycle correctly.
The diagram is divided into two sections indicated with a top and bottom horizontal line, and two vertical lines: one on the left and one in the middle. The horizontal lines intersect with the vertical lines. The left section has three horizontal lines intersecting with the left vertical line. Similarly, the right section has one horizontal line intersecting with the middle vertical line. There is one drop box at each intersection, and therefore, there are eight drop boxes.

The diagram is titled $Na^+(g) + Cl(g)$ mentioned at the top.
The left section features a vertical sequence of reactions with drop boxes arranged from top to bottom as follows:

$Na(g) + Cl(g)$
Dropbox
$Na(g) + ½Cl2(g)$
Dropbox
$Na(s) + ½Cl2(g)$
Dropbox
$NaCl(s)$

The right section features a vertical sequence of reactions and drop boxes arranged from top to bottom as follows:
Dropbox
$Na^+(g) + Cl^-(g)$
Dropbox

Users need to place the correct upward or downward arrows and the appropriate thermodynamic terms in the drop boxes provided in the cycle. The drag-and-drop options provided at the bottom from left to right include two rows of reads: Upward arrow, $\Delta H_f^\theta (NaCl)$, $EA(Cl)$, $IE(Na)$, Downward arrow, $\Delta H_{atom}^\theta (Na)$, $\Delta H_{lattice}^\theta$, and $E(½Cl2)$.
This interactivity helps users to construct and interpret the Born-Haber cycle for the formation of NaCl, demonstrating the enthalpy changes involved in the process.
Read below for the solution.
The arrows on the leftmost intersections from top to bottom in the table include upward arrow, downward arrow, upward arrow, downward arrow, and downward arrow. The arrows in the middle of the table from top to bottom include downward arrow, upward arrow, and downward arrow.
The reactions in the "Left section" from top to bottom are as follows:
$IE\ (Na)$
$E\ (½Cl2)$
$\Delta H_{atom}^\theta\ (Na)$
$\Delta H_f^\theta (NaCl)$
The reactions in the "Right section" from top to bottom are as follows:
$EA\ (Cl)$
$\Delta H_{lattice}^\theta$

The "Click" button on the bottom left allows users to check their answers. The number of correct answer(s) is indicated with a slider. When wrong answer(s) are input, the user can use the "Retry" button and redo the activity.

By following the directions of the arrows, we can see that there are two pathways to get to the step with the separated gaseous $Na^+(g)$ and $Cl^-(g)$.

Starting at the step $Na(s) + \frac{1}{2}Cl_2(g)$, the first pathway is equal to the enthalpy change of formation plus the lattice enthalpy ($\Delta H_f^\ominus + \Delta H^\ominus{}_{lattice}$).

The second pathway is equal to the enthalpy of atomisation, plus the bond dissociation energy, plus the first ionisation energy, plus the electron affinity ($\Delta H^\ominus{}_{atom} + E + IE + EA$). This can be represented in equation form as:

$$\Delta H_f^\ominus + \Delta H^\ominus{}_{lattice} = \Delta H^\ominus{}_{atom} + E + IE + EA$$

According to Hess's law, the enthalpy change for the two pathways is the same. By rearranging the equation, the lattice enthalpy can be calculated as shown:

$$\Delta H^\ominus{}_{lattice} = \Delta H^\ominus{}_{atom} + E + IE + EA - \Delta H_f^\ominus$$

Applied to the Born–Haber cycle in **Interactive 1**:

$\Delta H^\ominus{}_{lattice}(NaCl) = \Delta H^\ominus{}_{atom}(Na) + ½E(Cl_2) + IE(Na) + EA(Cl) - \Delta H_f^\ominus(NaCl)$

$\Delta H^\ominus{}_{lattice}(NaCl) = +110 + 121 + 496 + (-349) - (-411)$

$\Delta H^\ominus{}_{lattice}(NaCl) = +789\ kJ\ mol^{-1}$

Note that in this Born-Haber cycle, the bond enthalpy value E for Cl_2 is halved – this is because the formation of one mole of NaCl only requires one mole of chlorine atoms.

Worked example 1

Use the Born-Haber cycle to calculate the lattice enthalpy for potassium bromide.

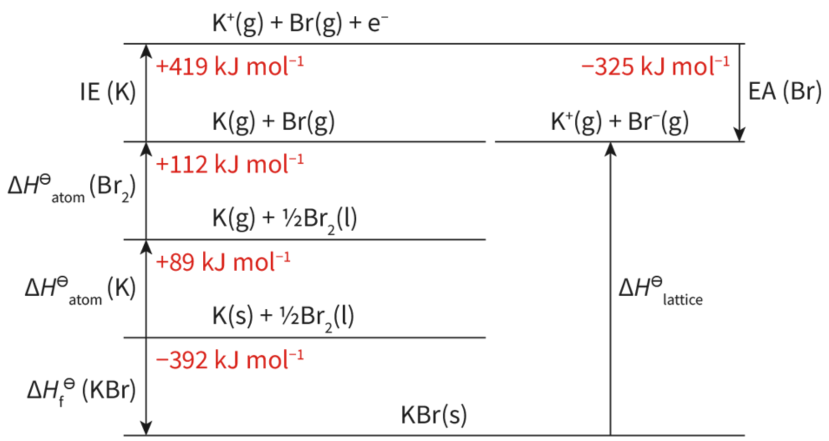


Figure 3. Born-Haber cycle for potassium bromide.

More information for figure 3

The image is a diagram representing the Born-Haber cycle for potassium bromide (KBr). It depicts a series of horizontal lines with arrows indicating energy changes during the formation of KBr. Starting from the bottom:

1. K(s) + ½Br₂(l) with an energy change of ΔH_f° (KBr) = -392 kJ mol⁻¹, moving to K(g) + ½Br₂(l) with ΔH°_atom (K) = +89 kJ mol⁻¹.
2. Moving upwards, K(g) + ½Br₂(l) becomes K(g) + Br(g) with an energy change +112 kJ mol⁻¹.
3. This progresses to K⁺(g) + Br(g) with an ionization energy change of IE (K) = +419 kJ mol⁻¹.
4. Finally, it transitions to K⁺(g) + Br(g) + e⁻ and an electron affinity change of EA (Br) = -325 kJ mol⁻¹.
5. The final step shows the formation of KBr(s), emphasizing the lattice energy ΔH°_lattice.

The diagram includes the chemical symbols, different energy changes (in kJ/mol), and states of matter (solid, liquid, gas).

[Generated by AI]

$$\Delta H^{\ominus}_{\text{lattice}} = \Delta H^{\ominus}_{\text{atom}}\ (\text{K}) + \Delta H^{\ominus}_{\text{atom}}\ (\text{Br}_2) + \text{IE} + \text{EA} - \Delta H_f^{\ominus}$$

Therefore:

$$\begin{aligned}\Delta H^{\ominus}_{\text{lattice}} &= 89 + 112 + 419 - 325 - (-392) \\ &= 687\ \text{kJ mol}^{-1}\end{aligned}$$

Worked example 2

Use the Born-Haber cycle to calculate the enthalpy of formation for calcium oxide.

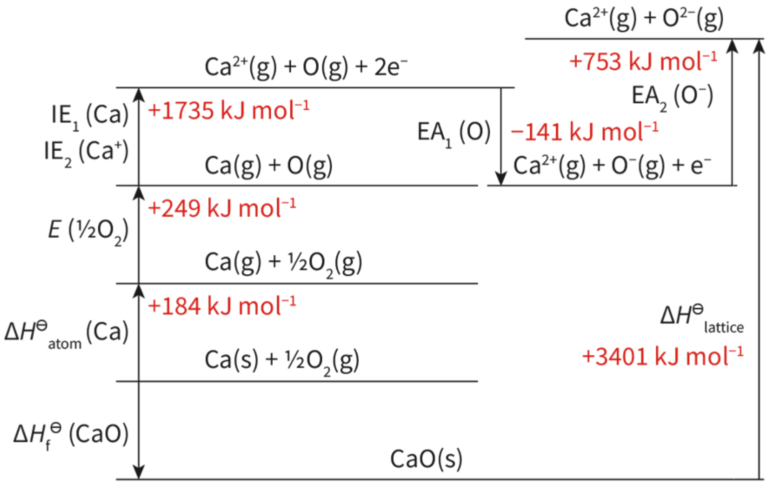


Figure 4. Born-Haber cycle for calcium oxide.

More information for figure 4

The image depicts a Born-Haber cycle diagram for the formation of calcium oxide (CaO). It illustrates various enthalpy changes, each accompanied by their respective equations and energy values in kilojoules per mole (kJ/mol). The cycle transitions through several stages: starting from the formation of gaseous calcium ions (Ca^{2+}) and gaseous oxygen ions (O^{2-}) at the top, moving through intermediate stages that involve ionization energies (IE), electron affinities (EA), and other enthalpies (such as atomization and lattice enthalpies). The cycle is presented in a series of horizontal and vertical lines, each level representing a specific chemical reaction or physical change, labeled with the corresponding enthalpy, such as +1735 kJ/mol for the ionization energy of Ca and +3401 kJ/mol for the lattice enthalpy of CaO. The diagram visually and textually explains the stepwise formation of CaO by illustrating the various energy changes involved.

[Generated by AI]

$$\Delta H^\ominus_{lattice} = \Delta H^\ominus_{atom} + E + IE + EA - \Delta H_f^\ominus$$

Therefore:

$$\begin{aligned}\Delta H_f^\ominus &= \Delta H^\ominus_{atom} + E + IE + EA - \Delta H^\ominus_{lattice}\\ &= 184 + 249 + 590 + 1145 + 753 - 141 - 3401\\ &= -621\ \text{kJ mol}^{-1}\end{aligned}$$

Worked example 3

Use the Born-Haber cycle to calculate the enthalpy of formation, ΔH_f, for sodium oxide.

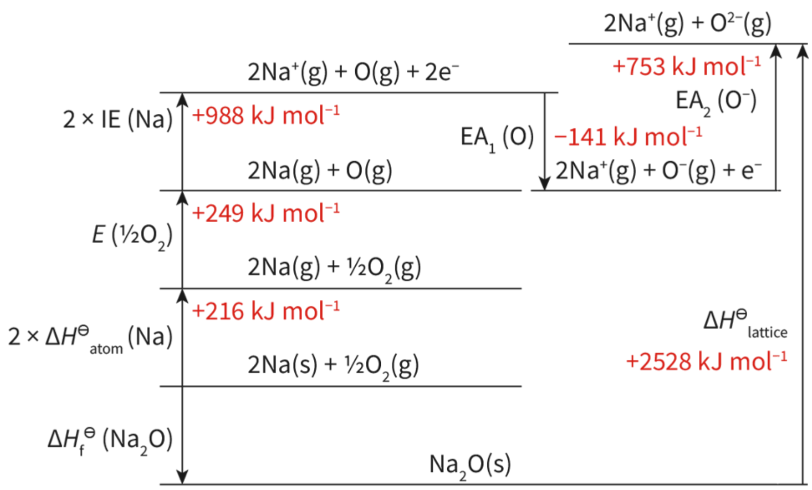


Figure 5. Born-Haber cycle for sodium oxide

More information for figure 5

The Born-Haber cycle diagram illustrates the series of steps involved in the formation of sodium oxide (Na_2O) from its elements. It includes the following stages:

1. The process starts with 2 moles of solid sodium (Na) and half a mole of oxygen gas (O_2), which is represented as (2Na(s) + \frac{1}{2}O_2(g)) transitioning to 2 moles of gaseous sodium atoms (2Na(g) + \frac{1}{2}O_2(g)), with an enthalpy change of +216 kJ mol^{-1} labeled as (2 \times \Delta H^\circ_{atom} (Na)).
2. This is followed by the excitation to sodium ions and oxygen atoms (2Na^+(g) + O(g) + 2e^-) with an ionization energy of +988 kJ mol^{-1} labeled as (2 \times IE (Na)).
3. Next, the process involves the conversion of oxygen atoms to form negative ions (2Na^+(g) + O^-(g) + e^-) with an electron affinity of -141 kJ mol^{-1} labeled as (EA_1 (O)).
4. This is followed by further electron gain (2Na^+(g) + O^{2-}(g)) with a second electron affinity of +753 kJ mol^{-1} labeled as (EA_2 (O^-)).
5. The final step is the formation of the ionic lattice (\Delta H^\circ_{lattice} = +2528 kJ mol^{-1}).

The overall enthalpy of formation ((\Delta H^\circ_f (Na_2O))) is derived by summing these steps together to yield the enthalpy change for the process.

[Generated by AI]

$$\Delta H^{\ominus}{}_{\text{lattice}} = \Delta H^{\ominus}{}_{\text{atom}} + E + \text{IE} + \text{EA} - \Delta H_f^{\ominus}$$

Therefore:

$$\begin{aligned}\Delta H_f^{\ominus} &= \Delta H^{\ominus}{}_{\text{atom}} + E + \text{IE} + \text{EA} - \Delta H^{\ominus}{}_{\text{lattice}}\\ &= 216 + 249 + 988 + 753 - 141 - 2528\\ &= -463\ \text{kJ mol}^{-1}\end{aligned}$$

Factors that affect the lattice enthalpy of an ionic compound

The magnitude of the lattice enthalpy of an ionic compound depends on two factors; the charge on the ions and the radius of the ions. The greater the charge and the smaller the radius, the higher the value of the lattice enthalpy. The table below has data for two ionic compounds, NaCl and KCl.

Compound	Cation radius (× 10^{-12} m)	Anion radius (× 10^{-12} m)	$\Delta H^{\ominus}_{lattice}$
NaCl	102	181	790
KCl	138	181	720

From the table, we can see that the difference in the lattice enthalpy is a result of the difference in the cation radius. The sodium ion, Na^+, has a smaller radius than the potassium ion, K^+. This results in a stronger attraction between the ions and a higher lattice enthalpy value.

With regards to the effect of ionic charge, we can compare the lattice enthalpies of NaCl and $CaCl_2$, which are 790 kJ mol^{-1} and 2271 kJ mol^{-1} respectively. The higher lattice enthalpy of $CaCl_2$ is a result of the 2+ charge of the calcium ion, Ca^{2+}, compared to the 1+ charge of the sodium ion, Na^+.

Worked example 4

State and explain which ionic compound has the highest lattice enthalpy, LiF or RbF.

Li^+ has a smaller ionic radius than Rb^+ so LiF would have the highest lattice enthalpy.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills — Presenting data appropriately
- **Time required to complete activity:** 30 minutes
- **Activity type**: Individual/pair activity

Sometimes it is easier to retain knowledge when you are able to manipulate objects. For this activity, obtain a piece of cardstock or cardboard and write out the ‘puzzle pieces’ of the Born-Haber cycle.

- Use these puzzle pieces to create the Born-Haber cycles for the following compounds: LiCl, NaBr, MgO

- Use a pencil to write the equations for each part of the Born-Haber cycle on the cardboard.
- Shuffle the pieces and then rearrange them into the correct Born-Haber cycle. Challenge yourself by doing this without looking at your notes!
- Take a photo of each when you are done so that you can use the information for calculating the lattice enthalpy.